(iii) Consider another example $2.25 \div 0.9$

Now $2.25 \div 0.9 = \dfrac{2.25}{0.9}$

Since $2.25 = \dfrac{225}{100}$ and $0.9 = \dfrac{9}{10}$ we get,

$$\frac{2.25}{0.9} = \frac{\left(\frac{225}{100}\right)}{\left(\frac{9}{10}\right)} = \frac{225}{100} \times \frac{10}{9} = \frac{225}{9 \times 10} = \frac{225}{90} = 2.5.$$

Try these

Divide the following.

(i) $\dfrac{9.25}{0.25}$ (ii) $\dfrac{8.6}{4.3}$

(iii) $\dfrac{44.1}{0.21}$ (iv) $\dfrac{9.6}{1.2}$

Hence, we can conclude that when both dividend and divisor has same number of decimal digits the division is as simple as the usual division of two natural numbers. When dividend and divisor has different number of decimal digits, first change the divisor to a whole number by multiplying suitable powers of 10. Then multiply the dividend by the same powers of 10.

Example 1.25 A car covers 16.8 *km* in 0.21 hours. What is the distance covered by the car in one hour.

Solution

The distance covered by the car in one hour $= \dfrac{16.8}{0.21}$

Multiply the divisor by 100 to make it a whole number. Hence multiply the dividend also by 100.

Therefore, $\dfrac{16.8 \times 100}{0.21 \times 100} = \dfrac{1680}{21} = 80$.

Example 1.26 The perimeter of a regular polygon is 17.5 *cm*. Each side measures 2.5 *cm*. Find the number of sides of a polygon.

Solution

Since it is given that the polygon is a regular polygon, all sides will be equal.

Therefore, the number of sides of a polygon $= \dfrac{\text{perimeter}}{\text{one side}}$

$= \dfrac{17.5}{2.5} = \dfrac{175}{25} = 7$ (Equal number of decimal digits in both dividend and divisor)

Hence, the polygon has 7 sides.

Think

The price of a tablet strip containing 30 tablets is ₹ 22.63 Then how will you find the price of each tablet?



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